

# IVY MATOBORI

INSTRUMENTATION & CONTROL ENGINEER

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ivy-matobori

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## EDUCATION

Jomo Kenyatta University of Agriculture and Technology (JKUAT)

Nairobi, Kenya

Bachelor of Science in Control and Instrumentation Engineering

2019 – 2024

## TECHNICAL SKILLS

- Instrumentation & Control:** Sensor and actuator integration, calibration procedures, data acquisition, real-time process data logging, energy metering, closed-loop control
- Industrial Automation:** PLC programming, Siemens TIA Portal, HMI configuration, machine control, alarm handling, safety interlocks
- Industrial Communication:** Modbus RTU/TCP, industrial Ethernet, MQTT, RS485, CAN bus, UART, SPI, I<sup>2</sup>C, LoRa, BLE, 4G/LTE
- Embedded Systems & Firmware:** Embedded C/C++, FreeRTOS, ESP-IDF, ESP32, bare-metal development, peripheral drivers, MCU selection, low-power design, Embedded Linux (in progress)
- Electrical & PCB Design:** Altium Designer, KiCad, Fusion 360, mixed-signal and analog design, schematic capture, multi-layer PCB layout, wiring diagrams, control panel design, field wiring, protection device selection, DFM, PCBA, board bring-up
- Debugging & Validation:** Oscilloscope, logic analyzer, signal integrity, root cause analysis, design reviews, Git

## WORK EXPERIENCE

Quepay Limited, Kenya | C++ · FreeRTOS · ESP-IDF · Altium · LTE/BLE/I<sup>2</sup>C

Feb 2025 – Present

Hardware Designer & Technical Operations Lead

On-site, Nairobi, Kenya

- Sustained a **99%** device uptime SLA across **KES 10M+** in processed transactions, as measured by fleet monitoring, by building the C++ and **FreeRTOS** firmware that actuates volume-metered dispensing and streams telemetry over LTE.
- Consolidated **6+** machine verticals onto one controller board, from water and milk ATMs to vending and laundry, by designing the field-deployed hardware and **4-layer** PCB in **Altium Designer**, scaling deployment **4×** in **4 months**.
- Delivered **six** field station control system installations by specifying, procuring, installing, and commissioning controller and sensing hardware, completing on-site validation and achieving customer acceptance before operational handover.
- Developed **calibration and verification procedures** for flow and temperature sensors, holding milk ATM dispensing accuracy within **±1.8%** of setpoint, by troubleshooting instrument failures and driving long-term measurement repeatability.
- Improved system reliability by keeping dispense faults below **0.6%** across **250** field-deployed units through the implementation of fault detection algorithms, alarm handling, safety interlocks, over-dispense protection, and automatic payment reversal.
- Reduced field return rate **58%** across **250** units, by driving root cause analysis on **15** production failures and introducing board bring-up, first-article inspection and test procedures that cut per-unit validation to **12 minutes**, leading a **4-person** team.

Veno Autobotics, Kenya | Data Acquisition · Embedded Linux · Modbus/RS485

Apr 2024 – Jan 2025

Embedded Systems Engineer, Embedded Hardware Design

On-site, Nairobi, Kenya

- Designed the sensing hardware for Veno's Level Sensor, achieving measurement accuracy within **1.5%** over a **4 m** range across **12** field sites, verified by bench characterisation against a NIST-traceable reference standard.
- Implemented the data acquisition and telemetry architecture for the VeNode energy meter, integrating **eight** live channels per site into a **SCADA** and visualisation platform over MQTT, with remote diagnostics and automated threshold alarms.
- Integrated **five** sensing peripherals into an embedded Linux platform by developing kernel drivers and device tree configurations, while also implementing bare-metal C/C++ firmware for resource-constrained nodes.
- Cut field installation errors **43%** across **20** sites, by producing the electrical schematics, wiring diagrams and control panel layouts installers worked from, specifying terminal blocks, DIN-rail distribution and protection.
- Delivered **10+** **mixed-signal** hardware platforms from concept to production by owning requirements capture, schematic design, PCB development, prototype validation, and enclosure integration, partnering with mechanical and software engineers and mentoring junior talent.

Kenya Space Agency, Kenya | Power Electronics · Systems Engineering

Jan 2023 – Mar 2023

Satellite Systems Engineer, Electrical Power System (EPS)

On-site, Nairobi, Kenya

- Achieved a **5 W** orbit-average power budget at **92%** conversion efficiency, verified through bench characterisation, by taking a **6-layer** EPS power distribution board from schematic capture to fabrication release with DC-DC conversion, filtering and overcurrent protection on **4** redundant rails.
- Contained single-point failures to a single branch across **6** independently protected distribution paths, by engineering a fault-tolerant power architecture and selecting overcurrent protection and isolation devices for each branch.
- Cut standby current draw **18 mA** against the baseline design, by leading the EPS microcontroller trade study across **6** candidate parts scored on power, peripherals, reliability and radiation tolerance.
- Cleared **2** multidisciplinary design reviews with engineers from **5** national space agencies (Kenya, Uganda, Egypt, Nigeria, Rwanda), by producing the electrical interface, verification and test documentation the EPS design was assessed against.

## PROJECTS

Closed-Loop Process Control Rig | Siemens S7-1200 · TIA Portal · PID · HMI

2024

- Held tank level within **±1.4%** of setpoint under step disturbances, cutting settling time **37%** against the untuned baseline, by designing and tuning a PID process control strategy on a Siemens PLC in **TIA Portal**.
- Commissioned the experimental rig end to end, integrating **three** field devices over **4–20 mA** wiring and an HMI exposing live trends, manual override and alarm state to the operator.